

Dissertation Title:

**How can remembrance and death be explored and
experienced in digital narrative storytelling?**

Chuyao Wang

24009771

MSc Creative Computing

University of the Arts London: Creative Computing Institute

Supervisor: Matthew Jarvis

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Abstract

Death has been one of the most profound themes in human storytelling. In gaming, death represents failure and rebirth (Cuerdo & Melcer, 2020). In recent years, as game storytelling has continued to evolve and become more mature, themes such as death, memory, and existence have become a part of game design.

This immersion is created not just through visuals or interaction, but also through storytelling that allows gamers to feel as though they are a part of that immersion. This is because when storytelling is balanced well with gameplay, feelings of empathy as well as emotional investment in a game occur (Toh, 2023). This is according to Schäfer (2022) because when gamers make decisions or feel emotions through their characters, they become a part of a common identity that is shared with that character.

This research is centered on digital remembrance (and digital immortality) as it considers the ways in which narrative games use interaction and gameplay as a means of expressing loss as well as the continuation of life in digital form. This research will examine death and digital renewal in terms of digital storytelling.

To explore this theme, the study analyses a two-player narrative and cooperative game set within a digital realm, in which one player embodies Life and the other player embodies Death, together traversing a border region that separates two worlds. In this digital realm, death is transformed from an endpoint to a new form of digital memory.

1.Introduction

"Games can serve as spaces for meaning-making, allowing people to reframe loss and existence through virtual interaction."

—Eum & Doh, *A thematic analysis of bereaved adults' meaning-making experience of loss through playing video games* (2023)

1.1 Motivation

Humans are not able to evade death as a theme. It has always been something that artwork tries to realise and interpret. People recently addressed loss and existence in several ways, from ancient rituals to contemporary pieces of literature. This expression has expanded in the digital era. Memory and identity may come embedded as data thanks to technology. From social media memorials to digital avatars, humans start to "continue existing" through algorithms (Ziatdinov et al., 2024). Has a person nonetheless been viewed as "alive" when the body is left, but the data has already been present?

Such questions can still be addressed in depth, making use of video games. In contrast with books and movies, games show stories through choice and interaction. Players may experience them, and on top of that, reveal their own stories. As Toh (2023) writes, interactive storytelling "offer users the opportunity to enact stories rather than just watch them unfold onscreen."

The two main themes that this work explores, in researching how games may exhibit life, death, and digital rebirth, are narrative and immersion. The narrative can be transformed in the player's hands in narrative games, in which the experience will take on emotional and reflective dimensions. This paper will look into how interactive design may allow users to effectively view loss, memory, and the idea of remaining or existing in digital worlds through the merging of narrative theory and immersion theory.

1.2 Research Goals

This research discusses how digital immortality and death can be interpreted through narrative games. It stresses how immersive Storytelling allows users to come across digital Rebirth, loss, and memory. A two-player narrative game, with one player living and all the others dead, becomes the objective.

Players experience a feeling of connection after death as an immediate result of their cooperation and acknowledged memories. This study uses video games as an artistic language in order to examine existence, emotion, and continuity in the digital era.

2. Related Work

2.1 Narrative Games and Death Themes

Death generally seemed a retribution or challenge loop, something that moved players into the start of a level in early video games or abandoned health. Through repeated failure and retry, these designs granted players a feeling of accomplishment and skill.

As more indie games and narrative-driven designs started cropping up, more and more game designers began to rethink what death meant in games. Death is representative of a core theme of some books investigating loss, farewell, and continuation rather than failure. Based on a real family's story and loss, *That Dragon, Cancer* (Numinous Games, 2016), plays out grief and faith as their children break away. In *Spiritfarer* (Thunder Lotus Games, 2020), death seems depicted as a process of care and release. Boyd said that “*Spiritfarer* is an essential step forward in how games can create emotionally rich experiences”(Boyd, 2021; Toh, 2023).

What Remains of *Edith Finch* (Giant Sparrow, 2017) uses fragments of memory to explain the nature of life and times to its viewers.

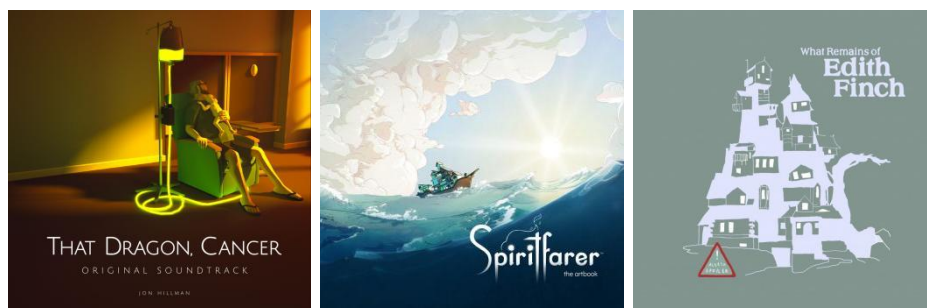


Figure 2.1: *That Dragon, Cancer*, *Spiritfarer* and *What Remains of Edith Finch*.

These pieces represent a shift from mechanical Storytelling into emotional Storytelling. Game play becomes an outlet to express emotions. Face loss, empathy, and meaning are now conveyed through play for players who no longer simply "control death". Boyd (2021) calls these "death-positive games," which discuss death gently and honestly, giving players emotional release and philosophical reflection in virtual space.

2.2 Death-Positive Games and Emotional Reflection

Boyd (2021) introduced the idea of death-positive games. These games do not ask players to avoid or fight death. Instead, they create a safe space for mourning and empathy through simple interaction.

For example, *A Mortician's Tale* (Laundry Bear Games, 2017) and *The Graveyard* (Tale of Tales, 2008) both encourage players to think about loss and find comfort in a calm virtual space.



Figure 2.2: From left to right, The Graveyard and A Mortician's Tale .

These works show that games can carry emotion and philosophy, not only entertainment.

My project follows this idea but goes further to explore digital continuation after death, turning reflection into connection between the living and the dead.

2.3 Immersion and Narrative Engagement

Important factors play a vital role in developing emotional experiences in video games. According to Toh (2023), interactive storytelling "offering users the opportunity to enact stories rather than just watch them unfold onscreen (Murray, 2017)." As a result, players become more mentally invested in the characters and experience deeper emotional engagement.

In addition, De Graaf et al. (2009) refer to immersion in the narrative world as "being out in the story world", where players exit the physical world and start the narrative.

For immersion of players, this project utilizes two-player cooperation as well as physical interaction. One player plays the deceased, and another plays the living, thus creating an aspect of cooperation for both players to achieve immersion in the story by progressing in the narrative as a result of combined actions by both players. Using this concept, a common story is achieved by the players, creating an emotional immersion between them as they work

together in either solving problems or progressing in the story by carrying out actions as per the story dictates.



Figure 2.3: Two-player interaction design showing emotional immersion.

2.4 Digital Immortality and Virtual Legacy

Personal identity and memory are no longer dependent on the body in the digital era. Social media accounts, photos, and messages easily remain online for a very long time after death.

Every rewind is an act of denial and remembrance at once. The player participates in this cycle of emotional repetition, reliving moments not to change them, but to understand them differently (Özgür Kalender, 2025).

The consequence of this situation is digital immortality, or in other words, the continuation of a person's data, memories, and identity after physical death (Ziatdinov et al., 2024).

Wang et al. (2024) created an interactive installation called Digital Cemetery. as a response to this idea. In this work, visitors can leave words, images, and digital traces in a virtual graveyard. Even though these traces exist, they are present for reading and interaction with the other visitors as so-called "digital tombstones." The more people extend the content, the more a digital "legacy" is born. This project thus visualizes that even if the body disappears, a person's digital identity and memories could be present.

My research is inspired by these ideas. It encompasses game telling with digital immortality. The project provides a living narrative experience by converting static data into a living narrative experience, where users may connect to virtualised versions of the dead. But does life persist in another way if thoughts and memories do live dynamically?



Figure 2.4: Digital Cemetery (Wang et al., 2024).

2.5 Research Gap and Direction

In both recent research work and video games, much attention has been paid to the role of death in narrative design and how immersion can facilitate emotional contemplation. This is explained in the work of Boyd (2021) and Toh (2023), for example, *That Dragon, Cancer*, and *A Mortician's Tale* are all part of a wave of death-positive games that encourage players to think about death and dying (Boyd, 2021; Toh, 2023).

But few studies, nevertheless, bring together the concepts of digital immortality, immersion, and death.

Through a narrative and interactive game that enables players to face loss and farewell, this project focuses on connecting these three elements. Use immersive interaction to instil social and emotional connection. Find out how to communicate one's identity and memory in virtual space using digital immortality. The combination of these will allow the research to open up new ways for video games to find life and continuity and fill a gap in current research.

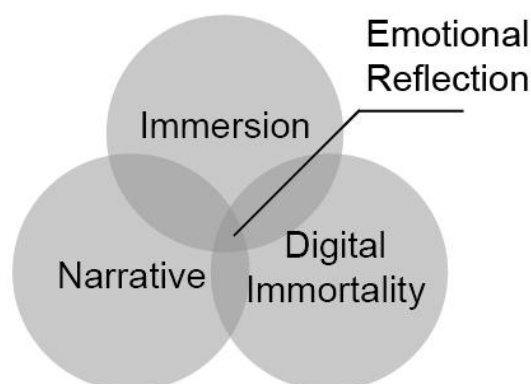


Figure 2.5: Research framework linking narrative, immersion, and digital immortality.

3. Methodology

This study uses a practice-based research approach. The primary strategy for creative making, where design and reflection are intertwined, is one of following ideas with meaning. Both the process leading to the research outcome and the conclusion form *the Bloom of Rebirth* project. The game provides a case for exploring how emotional and philosophical musings on digital immortality and death can be articulated through storytelling and engagement.

An iterative design process is followed wherein the game was designed, tested, and refined a number of times. Each version was enhanced through feedback and personal reflection. Such a process has helped understand how interaction and cooperation between players can build emotional connection.

The study is supported by a qualitative research method. Approximately ten players participated in session playtesting. Their reactions, comments, and short interviews were used to analyze emotional engagement and immersion. This data helped refine the story, balance the gameplay, and improve the emotional tone.

This combination of creative practice, user testing, and reflection evidences the nexus of making and thinking. By designing and playing, the research examines how life, death, and the continuation of memory can be represented through virtual space in digital games.

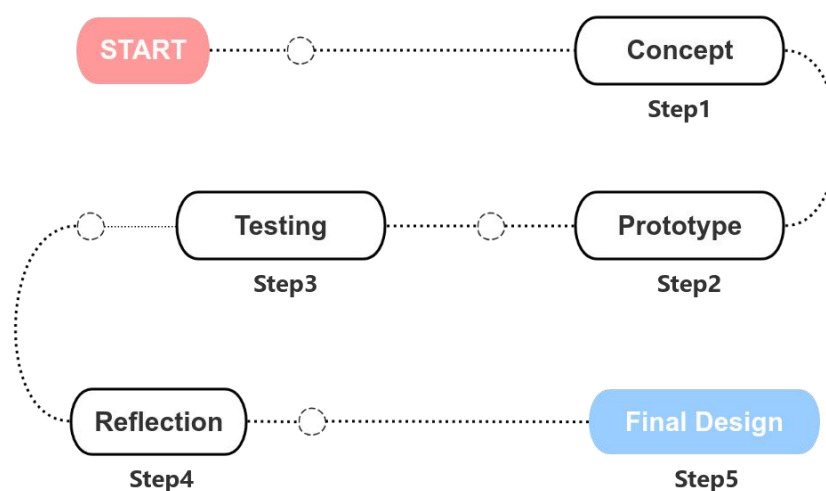


Figure 3.1: Research process.

4.A Project Write-Up

4.1 From Theory to Practice

This chapter moves from the theoretical framework of the narrative game to a real-life project known as *Bloom of Rebirth*. This project investigates how ritual, gesture, and space work as a barrier between souls of the deceased. This is a story that is passed through motion, rhythm, or light. This game is played as body motion that alters the environment through interaction with it. In this game, life and death become a shared experience for two participants.

In this installation, death is a reality of coexistence. There are two characters in the story—a living one and a dying one. They talk as they lie in two identical realms. When a character slides in one realm, it affects light or space in another. This is shown through memory and emotion that link both realms. Boyd (2021) said, "death-positive games use the concept of death to build empathy." In this context, empathy is a matter of connection, not loss. Language is replaced by gesture and rhythm, enabling a silent and gestural interaction between players.

The two performers work together to narrate a relationship between presence, absence, and connection. This work incorporates theories of collaborative design based upon embodied experience and immersion in storytelling. This research studies how digital gameplay can go beyond borders of bodies, states of being, and realms to craft an invisible emotional link through co-presence interaction.

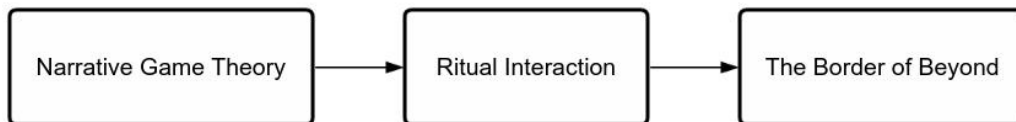


Figure 4.1. Conceptual framework: from narrative theory to interactive ritual design.

4.2 Literature Review and Theoretical Reference

Eum and Doh (2023) highlighted that video games have the ability to present “a unique meaning-making experience to bereaved players by affording them an agency to actively reconstruct their narrative of loss.” In relation to their study, interaction might have played a crucial part in recomposing a sense of purpose through gaming. The project’s conception relied upon an understanding of death as a form of transformation rather than an end.

Toh (2023) explained that “interactive stories” could be defined as “Interactive stories in the form of narrative games can be powerful agents of personal transformation by offering users...” that allowed users to become an act in a story instead of just watching it. This inspired the non-verbal storytelling within the game, where gesture and spatial feedback become means of expression.

Luo suggested in 2025, “Death-themed games address permanent death as experienced in real life, connecting players through digital grief stories and thus inspire reflections on death.” His framework of empathy and remembrance guided the emotional design of the project; light, sound, and motion replace dialogue to evoke connection among the players.

Put together, these works demonstrate that interaction and immersion are capable of communicating complex emotions. *Bloom of Rebirth* takes this a step further, translating an understanding of narrative and embodiment theory into a cooperative structure where the living and the dead share the same interactive world.

4.3 Narrative Translation into Design

This section will explain how *Bloom of Rebirth* turns narrative theory into a interactive experience.

4.3.1 The Game as Performance: From Player to Performer

Bloom of Rebirth is not a game meant to be passively watched—it is meant to be performed. Players engage with the narrative through bodily movement, rhythm, and spatial interaction. As Toh (2023) states, “the player experience is defined as the participants’ understanding of the game structure as a whole, which consists of the different relationships between narrative and gameplay in the ludo narrative model and contextual factors such as the player’s prior gaming experience.” In this respect, player actions become not only just a control scheme, but actually a means of storytelling in its own expressive language.

The game is a form of immersive performance, in which it communicates through light, sound, and motion instead of words. Every transition of an environment or lighting is a form of scene transition in a theater, creating an emotional context. In this game, death is not a theme but a backdrop—the force that defines a world. One player will represent “the Living,” while the “Dead” will be represented by the other player’s actions in each player’s world. Through this, death is given a new

form, not as an ending but as a bridge in which “life, memory, and presences” go through.



Figure 4.2. In-game performance moment.

4.3.2 Environmental Storytelling

In “*Bloom of Rebirth*,” the environment is not just passive setting—the environment is actually the foundation of the whole game. “...death is an extant feature of the spatial architecture of the game, part of the process of ‘world-building’ ...”(Spokes 2018). In *Bloom of Rebirth*, environment designs include a mix of sculpted and static meshes, with Blueprints and Niagara systems that respond dynamically to actions undertaken by the player. In essence, through lighting, fog, and particles, “the environment itself is a living part of the story.”

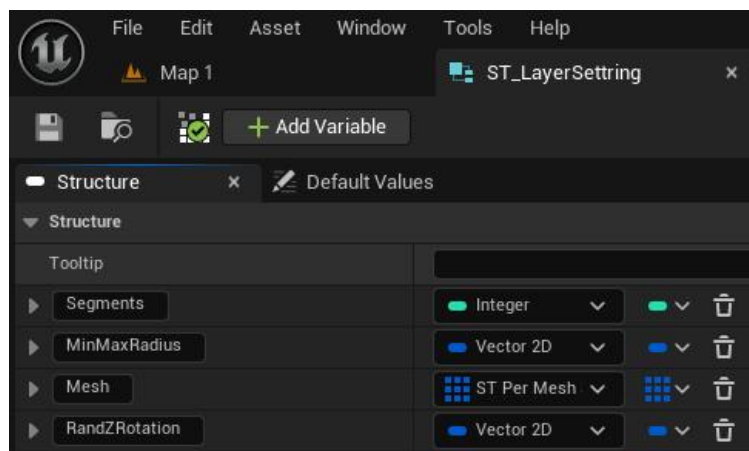


Figure 4.3. Landscape layer setting and mesh composition displaying terrain segmentation.

In the first stage, a warm offering area with marigolds and sugar skulls is entered by the player. Drifting petals, created through the use of the Niagara system NS_Chrysanthemum, represent memory and spirits. Triggered by a Trigger Volume in front of the offering area, a Timeline is initiated that increases light intensity and glow while spilling incense. Through this effect of lighting and motion, a physical environment is transformed into a ritual place.

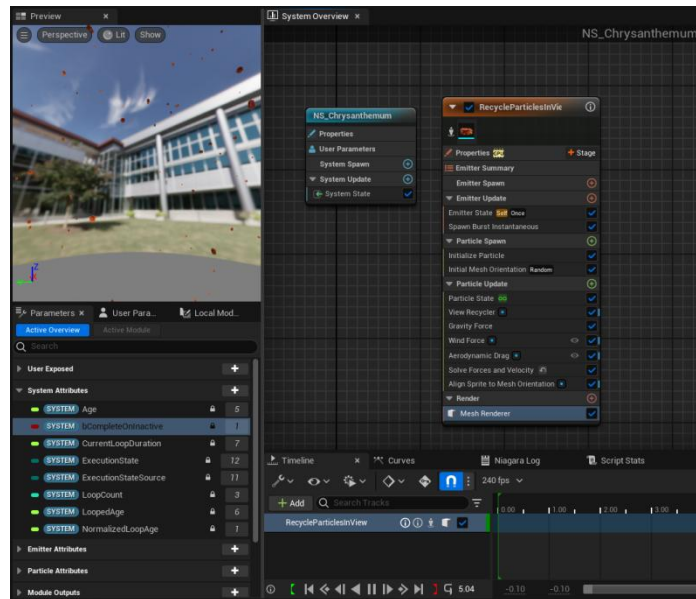


Figure 4.4. NS_Chrysanthemum.

In “The Second Level,” a cyber-like “Time Corridor” is created. By employing “NS_Fog” and lighting techniques that make use of contrasts, a state of liminality is created—one that is in a realm between death and life. “NS_Fog” is animated in terms of its density through a “Timeline” depending upon player motion. On one hand, “NS_Skylantern” produces “lanterns” moving at a “linear velocity” while “rotating randomly,” illustrating wishes and memories that climb higher in the sky.

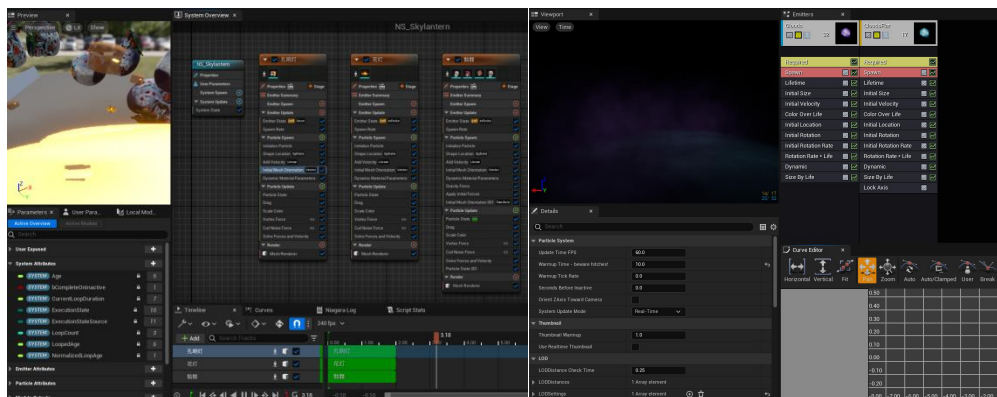


Figure 4.5. NS_Fog and NS_Skylantern.

In the third level, a Spline Blueprint is utilized for creating paths for the whale, as well as a river of light. This river is represented through “NS_Ghostparticle,” in which a skeletal mesh is involved in creating a silhouette of ghost-like objects that seem to drift. Within this setting, as an action is triggered involving a jump in a river, a “lerp” node is applied in order to make a transition from a fog environment to a bright night sky.

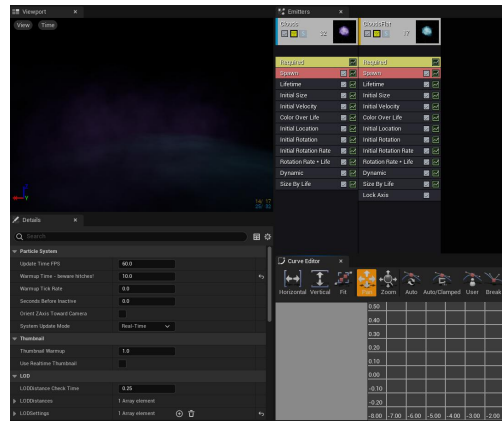


Figure 4.6. NS_Ghostparticle.

This is a system that incorporates spatial design, visual feed-back, and symbolic interaction. Every action that is performed by the player informs the environment, producing an embodied cycle of storytelling. “Environment storytelling is a means of telling a story through space,” said Jenkins in 2004. A system of language is created through space. This is through light, layers of fog, or a drifting petal that speaks of ‘life’ and ‘death’.



Figure 4.7. From left to right: Level 1, Level 2, Level 3.

4.3.3 Dual-player Structure and Emotional Connection

But at its heart is a two-player game system. This game is not just about death, but also about how those that are living relate to those that are deceased. One player would only be able to represent one of those sides of the equation, while

with two players, they are able to represent both sides. One player will represent those that are living, while the other player will represent those that are deceased. This allows death to be a much more tangible experience for those that are playing it.

Both characters affect each other in terms of light, space, and sound. Whenever the living character moves or produces light, an effect is triggered in the environment of the dead character. Similarly, whenever the dead character engages with the floating spirits, an effect is seen in the light and/or fog in the land of the living. Boyd (2021) writes that “... been extensive research in game studies about the qualities of empathy, embodiment, failure, and complacency found in video games, highlighting the medium's ability to make the interactor an active participant of what is happening on the screen, and evoking a meaningful response.” In this context, empathy is not delivered through dialogue, but emerges through cooperation and shared experience. This design also reflects Calleja’s (2011) “incorporation model,” which defines immersion as a situation where a player identifies through action.

In this manner, it is clear that the two-player system is more than just a gameplay system but also a storytelling element in which it is possible to have a story enacted through a collaborative process instead of simply being a passive experience. In this manner, it is possible to experience a sense of connection that goes beyond the realm of death and life.



Figure 4.8. Cooperative in-game.

4.4 Interactive and Technical Structure

Bloom of Rebirth is a combination of technology and storytelling. This game has a minimal dialogue system and is text-free when it comes to its storytelling aspect. Rather, all elements of storytelling are conveyed through “input, triggers,

and feedback.” This means that “the player’s body motion, positioning in space, and subjective experience of time become its language, driving the story.”

During the course of this experience, it encourages users to make particular actions that lead to a progression of events in a story. On completing such actions, it reacts in a technical manner such as a door slowly opening or a lighting effect that transports users through events as a part of a story at their own tempo.

4.4.1 Input and Gesture Recognition

Within “*Bloom of Rebirth*,” player body is utilized as a main interface and storytelling medium. A Kinect sensor is responsible for player motion detection, while TouchDesigner is responsible for transmitting it to Unreal Engine through an OSC system.

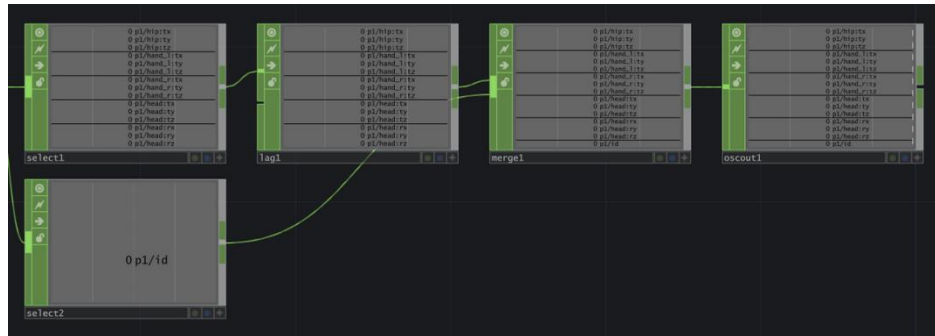


Figure 4.9. TouchDesigner to Unreal OSC Data Flow.

In Blueprint BP_Playerkinect, such signals are received and processed. Various nodes such as “Receive OSC Message,” “Get XYZ,” and “Make Vector” transform this data to work in world coordinates, which are stored in variables such as “lhead,” “lhand_L,” or “lhand_R.” This data is eventually utilized in other blueprints for controlling elements of a scene, such as door openings or particle systems, as well as modifying light intensity.

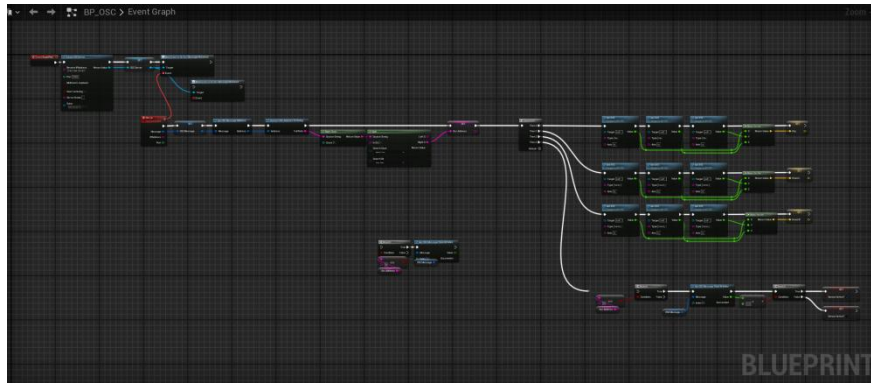


Figure 4.10. OSC Input.

When a player lifts a hand above a predetermined level or enters a specific region, a “Sphere Trace For Objects” function is used by the Blueprint to identify nearby objects that a player could act upon. Once an object is detected, a task event is triggered that will either attach or turn it on. This system is a means by which a player’s body motion is directly integrated as a story ingredient. By so doing, a gesture is more than an instruction—the entire motion is part of a game’s gestural vocabulary as a means of “speaking” through it.

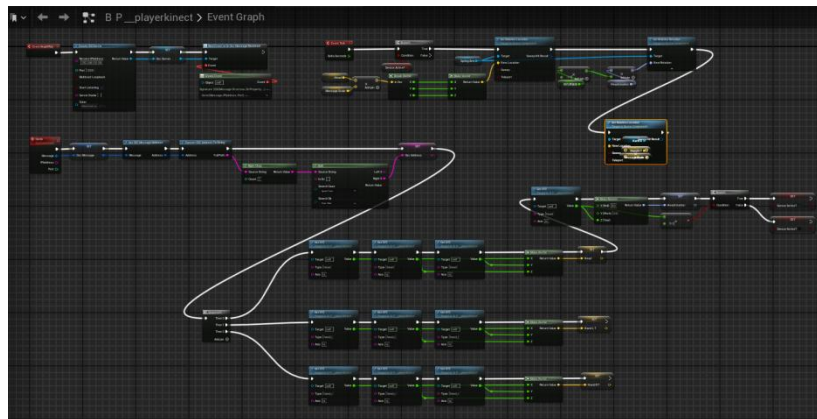


Figure 4.11. Get XYZ → Set Variable(BP_playerkinect).

4.4.2 Gesture–Event Blueprint Structure

Gestures serve as inputs as well as events that propel the narrative. The Blueprint system connects player physical behavior directly with interactions in a scene. Once the player’s hand is within a defined area, it employs “Sphere Trace For Objects” in order to identify nearby objects that can be interacted with. Based on the detection of an object in its environment, it performs actions such as “Attach Actor to Component,” “Set Visibility,” or “Activate Niagara Component.” This allows player body actions to influence the environment as well as drive the storyline.

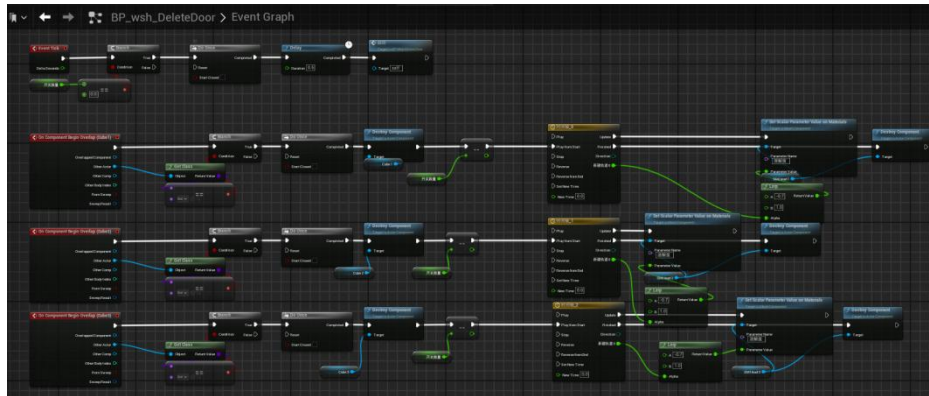


Figure 4.12. Door Interaction Blueprint (BP_wsh_DeleteDoor).

In the first stage, hand actions influence door rotation and lighting, which represents an entrance into memory. In the second stage, hand actions will help the character gather lamps. By utilizing “On Component Begin Overlap” and “Do Once” nodes, it is ensured that a particular event only triggers one action. Each of these events is integrated within a story structure, creating a continuous process of action, event, and feedback in which player action is a part of the overall story process in terms of its enactment in space.

4.4.3 System Integration and Emotional Response

In the installation *Bloom of Rebirth*, all of the Blueprints and data systems are integrated in order to form a whole interaction cycle. This is because a player's body actions are recorded by a Kinect sensor and sent through TouchDesigner in terms of OSC messages to Unreal Engine in real-time. In particular, within the engine, a BP_playerkinect Blueprint receives all of this data and passes crucial variables such as hand and head positions to various Blueprints, such as BP_AttachActor and BP_DeleteDoor. Once detected by Blueprints BP_AttachActor and BP_DeleteDoor, a player's gesture or action is followed by events such as door rotation, lighting changes, or object activation.

This creates a cycle that is all about input, detection, and feedback. This is because one player action changes the environment, and in turn, it reacts in such a way that it informs a player's actions in terms of what is supposed to happen next. This could include a light being turned on, a door opening, or a bunch of particles exploding—that is, it is not just a game mechanic but actually a reaction that tells a player that they are known by the game world.

“Body input, spatial feedback, and visual responses are integrated as a single system that converts technical designs into an emotional experience. Within such

a cycle, the player is tuned to the world—not just as a controller to dispatch commands but as a participant living within a story.”

4.4.4 Summary: Technology as a Narrative Medium

In terms of technology, in *Bloom of Rebirth*, technology is integrated in such a manner that it is completely absorbed in its storytelling aspect. There is a whole Blueprint system that ranges from BP_playerkinect to BP_DeleteDoor that is part of the storytelling system in such a manner that it is a component of code as well as an element of storytelling. Kinect and TouchDesigner bridge the gap between the player’s body and digital interaction, while Unreal engine converts it all into visuals that represent emotions.

This is where technical and emotional design come together in a synergistic manner. Rather than act as a passive viewer, the player is an active participant in telling that story. This is where code, sensors, and light intersect as a common language that facilitates immersion and empathy. Technology is not simply a means to an end in storytelling - it is the storytelling itself.

5.Evaluation

This chapter presents the evaluation of *Bloom of Rebirth* through a combination of practice-based reflection and user observation. In addition to creator-led iterative testing, fifteen participants were invited to interact with the prototype, each engaging in a structured fifteen-minute play session. The evaluation is organised into participant information, testing procedure, technical assessment, narrative assessment, and a quantitative summary.

5.1 Participants

A total of fifteen people took part in testing the *Bloom of Rebirth* for evaluation. Each person played the game for an average of fifteen minutes, resulting in a testing time of a cumulative total of 225 minutes. All testing took place face-to-face, where the researcher monitored gameplay mechanics to identify results of interactions, pauses in gameplay due to confusion, as well as emotional expressions.

5.2 Procedure

Each of the participants used a controlled testing environment to explore the Bloom of Rebirth prototype. The order of the study included instructions for them to freely play the game for a period of about fifteen minutes as they observed their gameplay actions. Interaction between the game mechanics, story structure, actions, pauses, pathways, and emotional responses of the study subjects was completed by the researcher during this time.

Item	Details
Number of Participants	15
Duration per Participant	15 min
Total Testing Time	225 minutes
Testing Method	On-site gameplay + researcher observation
Interviews Conducted	None
Data Collected	Behavioural observation, error logs, and qualitative feedback

Table 5.1 Participants Information Table

5.2.1 Technical Interaction Testing

The actions involved gesture-based fundamental operations such as opening doors, object activation, environmental response, which enabled analysis of

system delay, error detection in gesture recognition, as well as overall system stability in the integrated Kinect, TouchDesigner, Unreal Engine platform.

5.2.2 Narrative Comprehension Observation

The prototype's three-level structure was navigated by the subject. His capacity to grasp objectives, understand story-level guidance, recognize transitions, as well as his emotional involvement, was observed to assess the clarity of story coherence delivered by this prototype.

5.2.3 Post-Play Questionnaire

To complement behavioural observation, a short post-play questionnaire was distributed to all fifteen participants immediately after gameplay. The questionnaire served to capture players' subjective impressions of interaction clarity, narrative comprehension, emotional engagement, and the thematic expression of digital remembrance.



Figure 5.1. Game Experience & Interaction Feedback Survey

5.3 Technical Evaluation

The technological assessment has primarily focused on stability, responsiveness, as well as correctness, of an interaction system supporting *Bloom of Rebirth*. During fifteen playing instances, stability in performance of key interaction system elements, including body tracking using Kinect, OSC data transfer via TouchDesigner, as well as Unreal Engine Blueprints, has been achieved in synchronization.

5.3.1 System Performance

Testing revealed that this system supported real-time interactions in a stable manner. This is based on data from the OSC, which ranged from 0.12-0.15 seconds from TouchDesigner to Unreal Engine, without significant frames, freezes, and desynchronization of visual, audio, and physical feedback for critical actions in game development, such as level advancement, using all blueprint Blueprints including BP_PlayerKinect, BPAttachActor, and BPDeleteDoor.

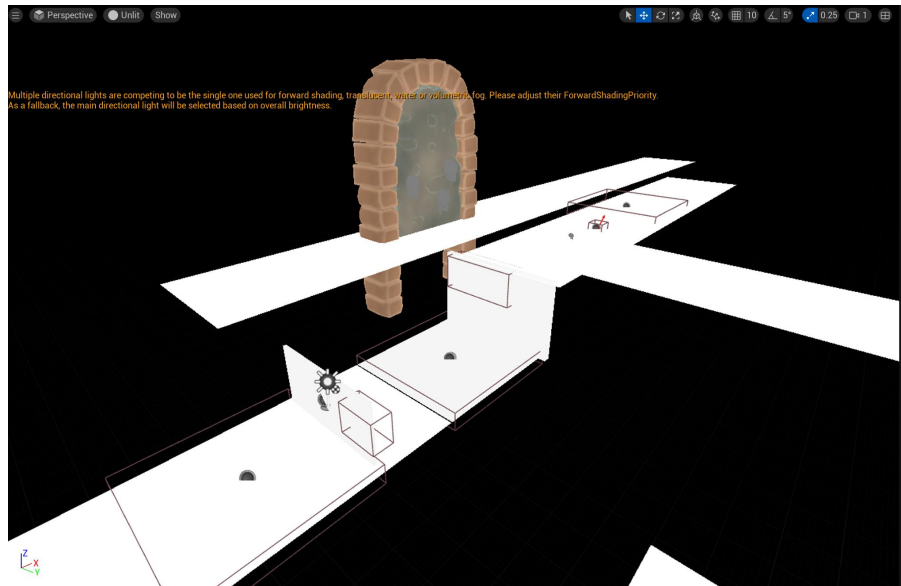


Figure 5.2 Collision Detection Test.

The gesture recognition precision is high, which enabled individuals to open doors, turn lights on, eliminate obstructions, and more without constant repetitions. Collision detection by Sphere Trace for Objects is also stable, identifying interaction zones without false triggers, even for gestured interactions that quickly repeat.

Technical	Test Focus	Result	Issues	Improvements
Data Latency	Kinect/TouchDesigner → Unreal OSC transmission	0.12–0.15 seconds	Minor delays in early builds	Adjusted recognition thresholds; optimised thread handling
Gesture Recognition	Trigger accuracy; false positives	Generally accurate with low error rate	7/15 participants experienced misreads under strong lighting	Added noise-reduction filters; adjusted lighting conditions
Blueprint Stability	BP_PlayerKinect, BP_AttachActor, BP_DeleteDoor integration	Stable and consistent	Earlier builds had incorrect event order	Reordered event execution and consolidated logic
Visual Effects	Light, particle, and environment feedback synchronization	Responsive and aligned with interactions	No major issues	-
Audio Triggering	Sound effect and ambient audio response	Synchronized with gameplay	No major issues	-
Collision Detection	Sphere Trace For Objects accuracy and trigger response	Reliable across all sessions	No incorrect triggers observed	-

Table 5.2 Technical Evaluation Metrics

5.3.2 Issues and Fixes

Even in the final build, many problems emerged in previous editions. The most critical issue is gesture recognition in bright lighting, which impacts 7 of 15 people. This issue is fixed by modifying recognition thresholds and implementing noise filters in TouchDesigner.

Secondly, there was an issue of Unreal Engine taking time to execute an event occasionally, causing a pause in interactions. Rearranging Blueprints to isolate high priority events corrected this issue, as after that, no pauses occurred at all.

Extended testing also vindicated system stability, with no loss of frames, memory leakage, or degradation of performance, which established that the technological framework is stable enough to support embodied narrative needs of this project.

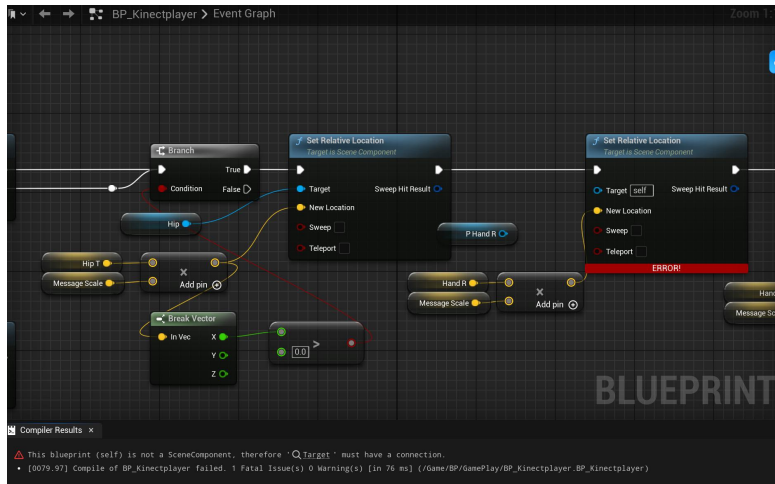


Figure 5.3 Blueprint Error Fix.

5.4 Storytelling Evaluation

The storytelling evaluation focused on clarity in gameplay regarding the narrative structure, gameplay, and emotional storytelling of *Bloom of Rebirth*.

5.4.1 Player Comprehension

The early testing showed that many interactions in the story did not get across to players without further support. Twelve of the fifteen players did not understand what they needed to accomplish in level one when they did not provide guidance in any form of speech bubbles or texts for that matter. Nine out of fifteen also experienced jarring transitions between levels as far as emotional tone, meaning they did not feel connected in those transitions either. In Level 2, six participants indicated a lack of motivation due to the absence of clear goals, and four participants noted that the original ending did not communicate the theme of rebirth effectively.

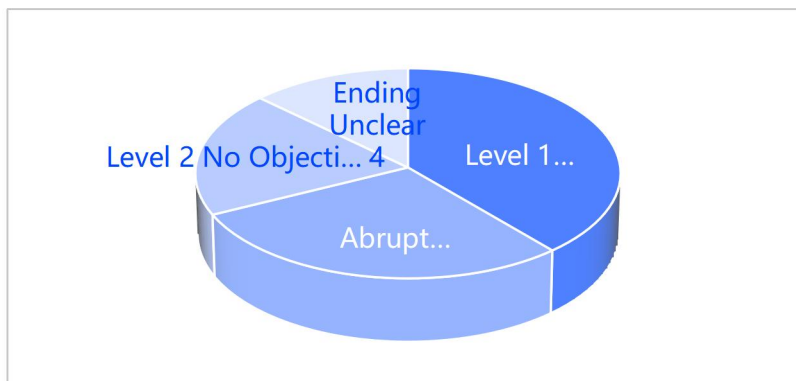


Figure 5.4 Narrative Issues.

These indicated a need for more narrative support in order to better allow player understanding in the more visually-based approach of Version One.

Category	Participants	Example Comments	Design Impact
Narrative	12/15 participants	"I don't know what I'm supposed to do."	Added NPC dialogue system and item-triggered story cues to guide exploration
Transition	9/15 participants	"It feels like switching to another game."	Introduced CG transitions and unified visual tone between levels
Objectives	6/15 participants	"There is no goal, so it feels meaningless."	Implemented mission objectives and completion feedback
Ending	4/15 participants	"I didn't understand the rebirth message."	Added final CG sequence and text to reinforce thematic closure

Table 5.3 Player Feedback Summary.

5.4.2 Iterative Improvements

As a result of problems that emerged, several refinements ensued. For early game confusion, an NPC conversation system as well as story triggers based upon items was incorporated into Level 1.



Figure 5.5 NPC Dialogue and Story Trigger System.

For dealing with discontinuity issues brought by sudden level transitions, cut-scenes as well as visual bridges between levels in CG games were introduced to smoothly shape emotional pace transitions from scene to scene.



Figure 5.6 Level Transition and CG Sequence

Finally, the ending is also refixed to include an actual CG sequence, as well as closing texts in order to provide more clarity to the message of birth by having a better organized ending.

Level 1	No text guidance; players could not understand objectives.	Added NPC dialogue system and item-triggered story events.	Players understood the gameplay faster; story flow became smoother.
Level Transitions	Disconnection between levels; emotional breaks in pacing.	Added transition animations and CG cutscenes with consistent tone.	Improved narrative continuity and emotional coherence.
Level 2	No clear completion condition; low motivation.	Added mission objectives and completion feedback.	Players felt more engaged and goal-driven.
Level3	Ending unclear; theme of rebirth not delivered.	Added final CG and short text for narrative closure.	Theme of “rebirth” became clearer and emotionally complete.

Table 5.4 Narrative Issues and Iterations

Collectively, these updates resulted in improved narrative comprehension, smoother emotional rhythm, and a more cohesive player experience throughout the game.

5.5 Quantitative Summary

Fifteen sets of quantitative results give a brief insight into the performance of the system for this study’s purposes. The interaction framework showed good stability by completing all necessary tasks, as well as maintaining an OSC latency of 0.12 to 0.15 seconds for all fifteen participants. Gesture recognition is

a reliable system, but 7 out of fifteen showed incorrect recognition in bright lighting conditions for earlier versions of the system.

The results of narrative comprehension showed significant improvement after several upgrades. In the first prototype, only 12 players could comprehend Level 1, while another 9 experienced problems in transition. However, after implementing speech aid signals and CG sequences, players continued in a more coherent manner emotionally as well as in game flow/objectives, in addition to having an improved ending that also corrected issues for those who lacked understanding of the original ending, as raised by 4 players.

Overall Evaluation Results		
Dimension	Summary	Summary
Stability	High stability with no major errors	15/15 completed tasks; OSC latency 0.12 - 0.15 seconds
Accuracy	Generally accurate, affected by lighting	7/15 experienced misrecognition under strong light
Narrative	Initial version difficult to understand; improved after updates	12/15 confused in early build; clearer after adding guidance
Immersion	Emotional flow improved with transitions	9/15 reported smoother transitions
Theme	Stronger thematic clarity after final CG	4/15 initially “did not understand rebirth” ; improved after revisions

Table 5.5 Overall Evaluation Results

Overall, the quantitative results indicate that iterative refinement enhanced both system robustness and narrative coherence, improving the clarity, flow, and emotional impact of Bloom of Rebirth.

6. Discussion

6.1 Interpretation

Bloom of Rebirth's inherent value is in its redefine of death through engagement and experience. In this game, word use is replaced by body motion so that experience of death is through action, not viewing—the player's body is part of the narrative act of moving from death to life.

In this game, death is not failure or an ending, but a feeling that may be felt and repeated. Every restart is a form of symbolic resurrection, depicting fear of loss as well as a digital concept of endless existence. In this game's system of repetition, a new understanding of death in a digital era is promoted, as a memory that returns or is experienced again.

By means of this cycle between body and system, *Bloom of Rebirth* transforms death into a sensation. This game encourages users to ponder existence as well as disappearance, figuring death as a part of digital existence as opposed to its opposite.

6.2 Relation to Theory

In a broad sense of creative research, death representation is often considered as a means for exploration of memory and existence. Normally, in traditional art, death is expressed either through symbols or through a story of remembrance, while in "*Bloom of Rebirth*," death is transformed into an experience that one may have through interaction. Besides telling a story of death, "*Bloom of Rebirth*" allows one to have an experience of death through action, thereby linking with Ryan's (2001) theory of experiential reality in that one is no longer a viewer but a participant in creating one's own reality.

Regarding its role as a medium, this is also an experience of incorporation in terms of how "the player's body and digital self become one" in gameplay, according to Calleja (2011). "Each gesture, pause, and repetition is incorporated in a form of narrative speaking," while it is also a "memory container system" as a whole. In effect, "memory recalls" in terms of lights, sounds, and spaces "save" death so it "can be remembered in a digital form."

This also evokes Becker's (1973) theory of *The Denial of Death*, that humans respond through symbolic action to extend self and mitigate death. Also, through its action/again cycle, it also brings out that death is transformation and not terminus.

In sum, “*Bloom of Rebirth*” departs from purely displaying death in a performance. Rather, it allows one to experience death through their own actions. This project assumes that “experiencing death” is a form of creative research in itself, in which one’s body and digital system cooperates in order to understand “life, memory, and continuation.”

6.3 Reflection and Future Work

Looking back at how it was done, *Bloom of Rebirth* is a good demonstration of both the power and limitations of body interaction gameplay. The use of body motion was beneficial in a sense that it led to immersion, though it also necessitated precise and concentrated motion, which was a challenge to execute correctly.

A possible challenge in body interaction gameplay is ensuring that control remains intuitive without disrupting immersion in gameplay.

Future research will concentrate on improved sensitivity and accessibility. Pose tracking using AI or machine learning calibration might make gesture recognition more flexible for various players. On the other hand, incorporating haptic or environmental components such as vibration and subtle changes in temperature might also aid in immersion. Finally, it could also be applied in installation contexts in which audiences collectively experience interaction as a type of performance or digital memorial.

The question that is asked is one of concept: how technology relates to emotion and how one’s presence in the world might continue through a digital form. In bringing performance and code together, *Bloom of Rebirth* expresses interaction through a poetic form of language. “The challenge that lies ahead is one of improving this process and research in how one’s embodied experience might enhance one’s connection or one’s digital existence.”

6.4 Contribution and Significance

In conclusion, *Bloom of Rebirth* opens new avenues in regards to the experience of death in digital media through practice as art. In contrast to traditional narrative structures in storytelling, *Bloom of Rebirth* does not take a textual or narrative form when it comes to storytelling about birth and death. Rather, through actions that make use of embodiment, it allows players to experience in a cycle of disappearing and becoming as possible through death in digital form, as an existence that is in a non-ending experience of death.

In terms of creative computing, it is clear that this project illustrates how technology might be used as an emotional storytelling medium. This project challenges technology as a purely technical object by making algorithm, delay, and system responses a part of its storytelling vocabulary. In effect, technology and emotion are treated simultaneously in one space, allowing a conversation between technology and emotion. This is basically an alternative approach to interactive storytelling based on Research Through Making.

In a broader cultural perspective, *Bloom of Rebirth* is a reflection of how human existence copes with death and memory in a digital era. This work challenges perceptions that technology is capable of preserving not only data but also emotion and a remnant of existence in a digital form when it allows users to “experience death in a digital world.” This leads to an open question of how human memory and existence will continue when death is something that is experienced and replayed. This is the most open research value of the project.

7. Conclusion

In this project, I explored whether death, memory, and continued existence via digital technologies could be communicated through embodied interaction rather than traditional storytelling methods. Prototype testing with 15 participants showed that gesture, repetition, and spatial transition can operate as narrative forms. Although only 46.67% of participants were familiar with narrative games and 100% had never played a death- or grief-themed game, most still interpreted death as transformation rather than failure. Early iterations revealed gaps in comprehension—only 100% understood the gesture–trigger relationship, 13.33% understood level transitions, and 0% understood the initial ending—yet embodied action nonetheless generated strong affective engagement, with 100% reporting emotional immersion and 93.33% finding cooperation meaningful.

By combining body interaction with Kinect technology, spatial storytelling, and co-operation with a second player, the project opens a conversation between technology and emotion. Death is represented not as a terminal or penalized process but as one of transformation—a moment of remembrance mediating between presence and absence. This aligns with Spokes’s (2018) concept of “environmental death as part of the spatial architecture of digital worlds”, and with Kalender’s (2025) view of mourning as “a cycle of denial and remembrance.” The practice extends these interpretations by translating interaction itself into a memorial language.

The results indicate that Bloom of Rebirth succeeds as a closed conceptual system in which meaning is produced through interaction loops, latency, and bodily engagement. The data-driven refinements—such as improved transitions, which raised narrative clarity to 100%—demonstrate how emotional rhythm and understanding can emerge from spatial and embodied cues rather than linguistic instruction. This positions technology as a narrative actor and affirms practice-based research as a method through which making becomes a form of theory.

8.List of Illustrations

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Figure 2.4: Digital Cemetery (Wang et al., 2024).

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Figure 4.3. Landscape layer setting and mesh composition displaying terrain segmentation.

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Figure 4.12. Door Interaction Blueprint (BP_wsh_DeleteDoor).

Figure 5.1. Game Experience & Interaction Feedback Survey

Figure 5.2 Collision Detection Test.

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Figure 5.6 Level Transition and CG Sequence

Table 5.1 Participants Information Table

Table 5.2 Technical Evaluation Metrics.

Table 5.3 Player Feedback Summary.

Table 5.4 Narrative Issues and Iterations

Table 5.5 Overall Evaluation Results

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Appendix

AI Reflection: Dissertation Unit CCI

This form is based on the work by Dr Rupert Norfolk, Dr Denise Kwan, Dr Marsha Bradfield.

Did you use AI in the making of this submission?

☒ Yes

☐ No

If you used AI in your submission, please complete the following questions.

Tool(s) Used (tick all that apply):

Which tool or platform did you use?

☐ Google AI

☒ ChatGPT (translation included)

☐ Claude

☐ Grammarly

☐ DALL·E

☐ Other (please list)

Purpose of AI Use (tick all that apply):

Why have you chosen to use AI?:

☒ Critical friend

☐ Idea generation

☒ Research tool

- ☐ Planning tool
- ☐ Drafting
- ☒ Checking grammar
- ☒ Image creation
- ☐ Accessibility support
- ☐ Other (please list)

What the AI Suggested?

- ☐ Text
- ☐ Image
- ☒ Outline
- ☐ Code
- ☒ Suggestions
- ☐ Essay plan
- ☐ Other (please list)

What did you do with it:

In this particular project, my primary use of AI was in the area of visualization and experimental images in the early concept phase. I chose not to utilize AI in my writing process in this assignment. In my writing process, I employed AI in checking some less-commonly-used words in the English vocabulary, namely “remembrance” in order to verify the appropriate English vocabulary use. When I was running out of ideas, I would chat with the AI in order to trigger my ideas, considering the fact that I was working alone. The AI also helped me in fixing some technical problems in Unreal Engine 5 since some explanations were better than those on the internet.

Reflections on Accuracy:

The AI's answers were mostly correct in the technical explanations and word meanings, and it was helpful when I needed an explanation quickly. But the AI's creative answers didn't always capture my artistic intention, so I was careful using those. I considered all answers as references, and nothing was usable on its own.

Ethical / Environmental Reflection:

I was aware of the ethics involved, like avoiding plagiarism, the bias in the AI, and making sure the end product was the result of my own creativity. I also made sure that I only utilized the tools when I had to, in other words, I didn't depend on the tools all the time. When I submitted work in Unit 9, the role played by AI was mostly in the visuals.

Support for Learning / Accessibility:

The AI tools assisted me in keeping my work organized, particularly in the process where I learned the use of tools in the UE5 environment on my own. The tools minimized some barriers in the process by allowing an immediate explanation that would otherwise take long. The tools also did not create barriers since I was rather selective.

Impact on Creativity / Confidence:

The AI tool sometimes helped me generate fresh ideas through conversation, particularly when I was stuck and needed someone, or rather, something, to talk to. AI gave me the confidence to consider other possibilities, but it would never take the role of my own creativity. AI simply provided the tool that fosters creativity, such as the reminder that the base design should come from myself.

Final Thoughts / Learning:

In this experience, I discovered the application use of AI, using it instead of, rather than in replace of, my own thoughts and ideas. I have realized the effectiveness of AI in the process of visual discovery and technical knowledge acquisition, where creativity and directions are beyond the use of AI. In my future, I would use AI selectively and retain all the rights and controls in my hands.